

SEMESTER END EXAMINATIONS - AUGUST / SEPTEMBER 2023

Program	: B.E. - Artificial Intelligence and Machine Learning / Artificial Intelligence and Data Science	Semester	: IV
Course Name	: Design and Analysis of Algorithms	Max. Marks	: 100
Course Code	: AI43 / AD43	Duration	: 3 Hrs

Instructions to the Candidates:

- Answer one full question from each unit.

UNIT - I

- Define the asymptotic bounds for algorithms with an example. CO1 (06)
 - Prove the following with respect to Stable Matching Algorithm:- CO1 (07)
 - If m is free at some point in the execution of the algorithm, then there is a woman to whom he has not yet proposed.
 - The set S returned at termination is a perfect matching.
 - Given the following asymptotic functions prove that it belongs to CO1 (07)
$$O(n^2), O(n^3)$$

$$f(n)=6n^2+2n+2$$

$$f(n)=10n^2+6n+3$$
- Define the following with respect to Gale-Shapley algorithm:- CO1 (04)
 - Perfect Matching
 - Stable Matching
 - Instability
 - Best Valid Partner.
 - Illustrate Competitive facility problem with an example. CO1 (06)
 - Identify the stable matching set S for the following set of men and women using Gale-Shapley algorithm. CO1 (10)

Men's preference list

A	K	L	M	N	O	P	Q	R
B	L	M	N	P	O	Q	K	R
C	M	K	L	N	P	O	R	Q
D	R	Q	P	O	N	M	L	M
E	L	N	M	Q	R	K	O	P
F	P	Q	R	K	L	M	N	O
G	O	P	Q	R	K	L	N	M
H	M	N	P	O	R	S	K	L

Women's preference list

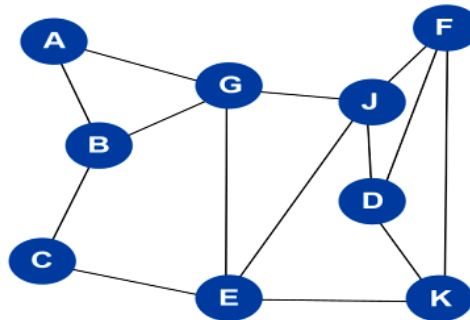
K	A	B	C	D	E	F	G	H
L	B	C	D	E	A	G	H	F
M	C	D	E	A	B	F	H	G
N	D	A	B	C	E	G	H	F
O	E	B	A	C	E	F	G	H
P	F	G	H	A	B	C	E	D
Q	B	C	D	E	A	F	G	H
R	E	F	G	H	B	A	C	B

UNIT - II

3. a) Identify the number of inversions for the array {9,10,3,4,8,2,15,3,2,1} CO2 (10)
using divide and conquer method.
b) Identify the BFS traversal for a graph represented with an adjacency CO2 (10)
matrix as follows:

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 0 & 1 \\ 1 & 0 & 1 & 0 & 1 & 1 \\ 0 & 1 & 0 & 1 & 1 & 1 \\ 1 & 0 & 1 & 0 & 1 & 1 \\ 1 & 0 & 1 & 1 & 0 & 0 \\ 0 & 1 & 1 & 1 & 0 & 0 \end{pmatrix}$$

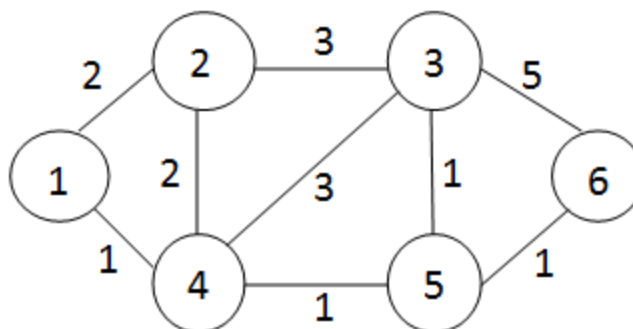
4. a) Write the BFS algorithm. Discuss its application in graph coloring CO2 (06)
problem.
b) Construct the DFS Traversal tree for the following graph CO2 (08)
with "A" as the source vertex and Write the algorithm for the same.



- c) Analyze the running time of Merge-Sort using CO2 (06)
i) Unrolling method ii) Substituting a solution.

UNIT - III

5. a) Prove that the greedy algorithm for the interval scheduling problem CO3 (06)
returns an optimal set of requests.
b) Identify the shortest path from source node 1 to all other nodes for the CO3 (08)
following graph using Dijkstra's algorithm.

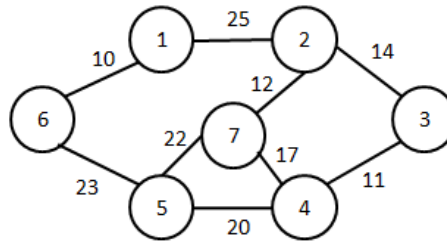


- c) Construct a Huffman tree for the following data: CO3 (06)

Char	A	B	C	D	E
Probability	0.4	0.1	0.2	0.15	0.15

- i) Encode the text A B A C A B A D using generated code.
ii) Decode the text whose encoding is 10001011100101.

6. a) Given the page frame=3 and page request 7 0 1 2 0 3 0 4 2 3 0 3 2 1 2 CO3 (08)
identify the number of cache miss using optimal cache miss algorithm.
b) Construct the minimum spanning tree for the following graph using CO3 (08)
Prim's algorithm.



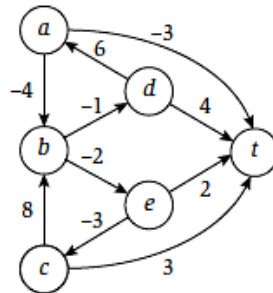
- c) Compare between greedy and divide and conquer techniques. CO3 (04)

UNIT- IV

7. a) Describe an algorithm to determine the shortest path for a graph with CO4 (06)
negative edges.
b) Describe an algorithm to determine the shortest path for a graph CO4 (06)
 $G=(V,E)$ with negative edge weights.
c) Identify the maximum subset-sum that can be achieved for the CO4 (08)
following items with maximum capacity $W=6$.

Item	Weight (W)
1	2
2	3
3	4

8. a) Describe survey-design problem for a given set of customers and CO4 (06)
products with an algorithm.
b) Assess bellman ford algorithm to find the [distance](#) from source CO4 (08)
[vertex 'a'](#) to destination vertex 't' in a [weighted digraph](#).



- c) Identify the maximum profit and items in the knapsack for $W=10$, CO4 (06)
 $(w_1, w_2, w_3, w_4)=(4, 7, 5, 3)$ and $(p_1, p_2, p_3, p_4)=(40, 42, 25, 12)$.

UNIT - V

9. a) Summarize Travelling salesman problem and construct a proof for its CO5 (06)
NP-completeness.
b) Describe the general strategy used for proving NP-complete problems CO5 (08)
with an example of Independent set.
c) Define NP complete problems and construct a proof for "Is there a CO5 (06)
problem in NP that does not belong to P? Does $P = NP$?"
10. a) Explain Polynomial time reduction with an example. CO5 (10)
b) Suppose you could reduce an NP complete problem to a polynomial time CO5 (10)
problem in polynomial time. Develop a solution for the consequence?
